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UNIVERSITÀ DEGLI STUDI DI MILANO
FACOLTÀ DI SCIENZE E TECNOLOGIE

Master's Degree in Computer Science

Investigating Ideological Bias in
Large Language Models:
A Comparative Study of Role and Language Effects

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Abstract

Contents

1	Introduction	3
1.1	Thesis Structure	4
2	State of the Art	5
2.1	Natural Language Processing (NLP)	5
2.2	Large Language Models (LLMs)	6
2.3	Bias in NLP and LLMs	7
2.4	Gap Analysis	8
3	Methodology	10
3.1	Data and preprocessing	10
3.2	Prompting and response collection	12
3.3	Analysis techniques and metrics	13
3.3.1	Agreement ratio	13
3.3.2	Embedding comparisons	14
4	Analysis of Results	15
4.1	Dataset Exploration	15
4.2	Topic Modeling Results	16
4.3	Agreement Ratio Analysis	18
4.4	Embedding Similarity Results	23
4.4.1	GPT-4o-mini	23
4.4.2	LLaMA 3.1-70B	24

4.4.3	General Patterns	24
4.5	Model Comparison (GPT vs LLaMA)	27
5	Limitations and Future works	28
	References	29

Chapter 1

Introduction

Large Language Models (LLMs) such as GPT [1] and LLaMA [2] are now deployed worldwide and used in many languages. As these systems mediate information access and public discourse, a central concern is whether they present the same opinions and reasoning to users across countries and languages—or whether they exhibit ideological or cultural biases [3, 4]. Understanding these effects is essential for trustworthy, fair, and transparent AI [5].

Existing audits have shown that LLMs can encode biases along several axes, including social, linguistic, and representational dimensions [6–8]. However, most empirical studies remain limited to a single country, a single demographic axis, or rely heavily on manual prompt probes and human judgments [9]. Less attention has been paid to cross-country and cross-language ideological behavior in contemporary chat models, and to disentangling the influence of role framing (e.g., “answer as a citizen of X”) from language framing (answering in X’s native language). Recent work has begun to examine political and national biases in LLMs [10–12], but reproducibility remains a challenge, since manual inspection is often subjective and difficult to replicate. This motivates the need for a multilingual and reproducible evaluation that targets both stance and reasoning on policy-relevant topics, while reducing dependence on ad-hoc human judging.

The objective of this thesis is to build such a framework. My approach is to evaluate ideological bias across roles and languages in a way that combines stance alignment with

semantic similarity of justifications [8]. In addition, I compare whether different model families—GPT and LLaMA—exhibit similar or diverging bias patterns, thereby situating model behavior within a broader comparative perspective [10]. This dual focus on stance and reasoning aims to provide a more fine-grained picture than agreement counts alone, while ensuring reproducibility.

Taken together, these motivations lead to four guiding research questions:

1. Do LLMs exhibit ideological bias across countries and policy topics?
2. Are model outputs more sensitive to role framing or to language framing?
3. How similar are role-based and native-language responses for the same country?
4. Are the observed effects consistent across model families (GPT vs. LLaMA)?

1.1 Thesis Structure

to be written in the end

Chapter 2

State of the Art

2.1 Natural Language Processing (NLP)

Natural Language Processing (NLP) is the field of computer science and artificial intelligence concerned with enabling machines to understand, generate, and interact with human language. Its applications range from information retrieval and machine translation to dialogue systems and text summarization. A central challenge in NLP is finding effective ways to represent linguistic units—words, sentences, or entire documents—in a form that computers can manipulate.

Early approaches to representation relied on static word embeddings such as Word2Vec and GloVe [13, 14], which mapped words to fixed vectors. These methods captured broad semantic relations (e.g., $king - man + woman = queen$) but could not account for polysemy, since each word type was assigned a single embedding regardless of context. The introduction of contextualized embeddings, with models like ELMo and BERT [15, 16], addressed this limitation by letting representations depend on surrounding words. This innovation provided richer semantics and led to major performance gains across a variety of NLP tasks.

Once words and sentences could be represented as vectors, similarity-based operations became possible. Sentence-level models such as SBERT [17] and cross-lingual encoders like LaBSE [18] extended these capabilities to semantic retrieval, clustering, and mul-

tiling alignment. Measures such as cosine similarity thus became central tools for comparing meaning, and they form the foundation of our later reasoning analysis.

2.2 Large Language Models (LLMs)

A language model is a statistical or neural system trained to predict and generate sequences of text. Earlier generations of neural language models were based on recurrent architectures such as RNNs and LSTMs, which captured sequential dependencies but struggled with long-range context and computational efficiency. These limitations motivated the search for architectures that could scale more effectively.

The introduction of the transformer architecture [19] marked a decisive shift. Its attention mechanism allowed models to process text in parallel and capture global dependencies, laying the foundation for today’s large-scale language models. The transformer framework can be adapted in different ways: encoder–decoder architectures such as T5, which combine a bidirectional encoder with an autoregressive decoder; encoder-only models such as BERT, optimized for understanding tasks; and decoder-only models such as GPT and LLaMA [2, 20, 21], designed primarily for text generation.

Training LLMs typically involves pretraining on massive text corpora using objectives such as next-token prediction or masked language modeling, followed by fine-tuning or instruction tuning to adapt the model to user-facing tasks [22]. While this paradigm enables strong zero- and few-shot generalization, it also makes LLMs highly dependent on the biases and imbalances present in their training data. As a result, stereotypes, ideological leanings, and cultural assumptions embedded in large-scale corpora can be learned and reproduced by the models, raising concerns about fairness and accountability [5, 23].

Today, LLMs are regarded not only as technical achievements but also as social actors, given their growing impact on communication, public discourse, and decision-making. Their training dynamics and inherited biases thus form a crucial part of understanding their broader societal implications.

2.3 Bias in NLP and LLMs

Bias in language technologies manifests in multiple ways. Social and representational biases perpetuate stereotypes around gender, race, or religion [7, 9, 24]. Ideological biases appear in politically sensitive contexts, for instance when models discuss international conflicts, sanctions, or freedoms [10, 12, 25]. Linguistic and cross-lingual biases arise from discrepancies between high- and low-resource languages, often disadvantaging less represented groups [11]. Efforts to measure these biases have evolved alongside NLP itself. Early work relied on embedding-level association tests such as WEAT and SEAT [6, 26, 27], followed by stereotype-focused datasets including CrowS-Pairs and StereoSet [7, 24]. In the era of LLMs, the focus has shifted toward prompt-based audits and open-ended evaluations [8, 25]. Our study builds on this trajectory by incorporating stance and justification into the analysis, moving beyond surface-level associations.

2.4 Gap Analysis

Table 2.1: Summary of representative related works, showing their focus, scope, and limitations relative to this thesis.

Study	Bias Type	Language(s)	Method	Limitations
StereoSet [7]	Stereotype (gender, race, profession, religion)	English	Association tests with contextual LMs	Focused on stereotypes; single language
CrowS-Pairs [24]	Stereotype (US-centric)	English	Paired-sentence probes	Narrow cultural scope
Gender Bias in ChatGPT [9]	Gender stereotypes	English	Prompt-based audit	Manual inspection; limited scope
OpinionGPT [25]	Political, demographic	English (Reddit)	Instruction- tuned model	Domain-specific; limited generality
Silence of the LLMs [10]	Political, misinforma- tion	Multilingual	Prompt-based, cross-lingual	Language effects not disentangled from role framing
Russia–Ukraine War [12]	Political neutrality	Chinese social media	Content analysis of LLM outputs	Single case; single conflict

As summarized in Table 2.1, prior research has concentrated on stereotype-centric audits in English, multilingual studies that do not clearly separate role framing from language

framing, and evaluation methods that emphasize associations rather than systematic comparisons of stance and reasoning. Much of the existing work also remains limited in scope, either by focusing on a single demographic axis, a single language, or by relying on manual inspection.

Our thesis addresses these gaps in three ways. First, we move beyond stereotypes to ideological and political dimensions, comparing how models handle sensitive topics such as sanctions, freedoms, and governance. Second, we explicitly disentangle role-based framing (e.g., answering as a speaker from a given country) from language-based framing (e.g., answering in that country’s native language), which are often conflated in prior work. Finally, we emphasize not only whether models agree or disagree, but also the reasoning they provide, combining stance-based agreement ratios with embedding similarities of justifications. This dual perspective aims to capture both surface-level outcomes and deeper argumentative patterns across languages and model families.

Chapter 3

Methodology

Our methodology follows the pipeline illustrated in Figure 3.1. We begin with a curated dataset of arguments, preprocess it through topic modeling and translation, and then prompt large language models under controlled conditions. Their outputs are collected, normalized, and evaluated through both stance agreement and embedding similarity. This section first introduces the data and its processing, then details the prompting and response collection procedure, and finally outlines the metrics used for analysis.

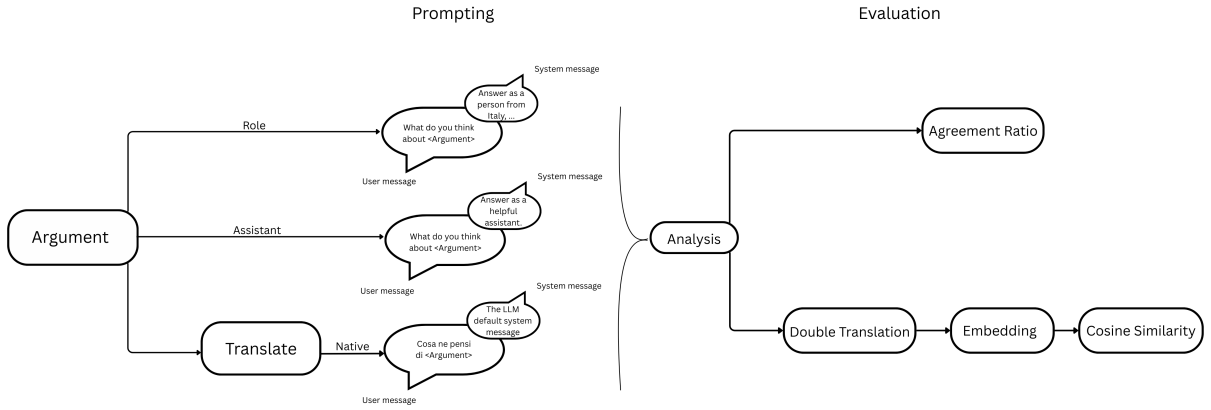


Figure 3.1: Methodology pipeline: BERTopic selection, prompting (role/native), stance agreement, and embedding similarity.

3.1 Data and preprocessing

We rely on the `arguments-training.tsv` file from SemEval 2023 Task 4: ValueEval [28], derived from the Webis-ArgValues-22 corpus [29]. It contains 5,393 arguments annotated

with a premise, a conclusion, and a stance label (*in favor of / against*). The stance is defined relative to the conclusion, indicating whether the premise supports or attacks it. Representative examples are shown in Table 3.1.

Argument ID	Conclusion	Stance	Premise
A01009	We should fight for the abolition of nuclear weapons	against	nuclear weapons help keep the peace in uncertain times.
A05040	We should legalize sex selection	against	we should not do this because other countries like China already did and it caused an epidemic where there is a huge imbalance of males to females which disrupts the population.
A05057	We should subsidize Wikipedia	against	subsidizing Wikipedia could be a back gateway to some form of governmental control, debasing the validity of the articles and information provided.
A05048	We should ban fast food	against	fast food is sometimes the only way poor people can feed their families cheaply. If you ban it then you are taking away someone’s only meal.
A05089	We should ban missionary work	against	missionary work is beneficial for those receiving aid.
E04002	We need to invest in nuclear energy	against	Nuclear energy is too expensive and hence economically not viable. According to researchers at RethinkX, a clean energy disruption in the next ten years may leave investments in oil, coal, and nuclear stranded.
E01030	Gas and nuclear power should be supported with money from European countries	against	Continuing to invest in environmentally damaging and immensely expensive nuclear power plants is a waste of resources that could be used for sustainable transformations.
A05017	Cosmetic surgery is unnecessary and dangerous.	in favor of	We should ban cosmetic surgery.

Table 3.1: Sample arguments with conclusions, stance labels, and premises.

Because the arguments span diverse and fine-grained issues, we apply BERTopic [30] to cluster them into broader themes. Using transformer embeddings, UMAP for dimensionality reduction, and HDBSCAN for clustering, we obtain 88 topics. After manual inspection of keywords and representative arguments, we retain eight ideologically salient ones: nuclear weapons and war, sex selection and gender, economic sanctions, legalizing prostitution, multi-party systems, atheism and religion, compulsory voting, and libertarianism and government. The distribution of arguments across these themes is shown

in Figure 3.2, ensuring that our evaluation centers on socially and politically sensitive debates.

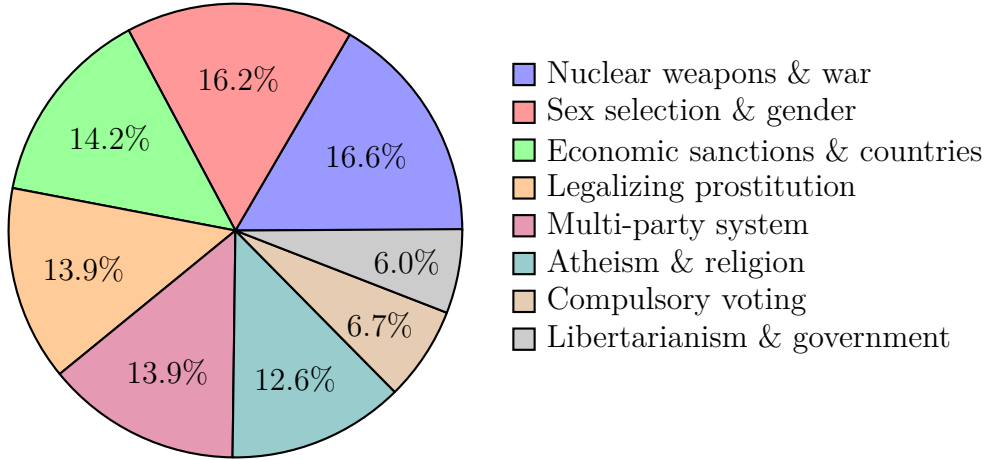


Figure 3.2: Topic distribution across selected themes.

3.2 Prompting and response collection

Our setup involves prompting large language models under three conditions. In the *country-role* condition, the model is instructed to answer as if speaking from a national persona (America, Iran, Israel, Ukraine, Germany, Italy, France, or China). In the *assistant-role* condition, the model responds as a neutral assistant without any national framing. Finally, in the *native-language* condition, the same instruction is provided in the country’s primary language, without an explicit persona. To prepare these prompts, we use the **deep-translator** Python library [31], which provides access to Google Translate for consistent translations across languages.

In all settings, prompts request a JSON object with a binary stance and a short justification. Where possible, we enforce strict JSON output using the API option `response_format={"type": "json_object"}`. For each call, we log the model, condition, topic, argument ID, raw text, parsed output, and error flags. This ensures a complete trace for reproducibility and error analysis.

Two model families are queried: OpenAI’s GPT-4o-mini, which supports native JSON enforcement [32], and Meta’s LLaMA 3.1-70B [2].

3.3 Analysis techniques and metrics

Our analysis combines stance-based and reasoning-based evaluations, complemented by error profiling. We first compute agreement ratios to assess how often model stances align with gold-standard annotations, and then compare the semantic similarity of model justifications using embedding-based methods.

3.3.1 Agreement ratio

Agreement ratios capture whether a model’s predicted stance on a premise implies agreement with the conclusion stance given in the dataset. For topic t and condition r , we define

$$\text{AgreeRatio}(t, r) = \frac{\# \text{agree}}{\# \text{agree} + \# \text{disagree}}.$$

The mapping between model stance and dataset stance is shown below:

Model stance	Argument stance	Model $\rightarrow$ Conclusion
Agree	In favor of	Agree
Disagree	In favor of	Disagree
Agree	Against	Disagree
Disagree	Against	Agree

For LLaMA outputs, we extend the denominator to handle **Neutral/Partial** cases:

$$\text{AgreeRatio}^{\text{LLAMA}}(t, r) = \frac{\# \text{agree}}{\# \text{agree} + \# \text{disagree} + \# \text{neutral}},$$

where **Partially Agree** is mapped to **agree**, and **Neutral/Neither** to **neutral**. Agreement is counted when the inferred stance matches the gold-standard conclusion stance, providing a robust stance-level measure across topics and conditions.

3.3.2 Embedding comparisons

While stance labels capture surface agreement, the justifications supplied by models provide richer insights into their reasoning. We embed these free-text reasons using LaBSE [18] and compute cosine similarities row-by-row between outputs from the three conditions: role-based, assistant-role, and native-language prompts. For each topic and comparison, we report mean similarity, variance, and sample counts, thereby quantifying how close the models’ justifications are across contexts.

A challenge in this setting is that embeddings produced directly from native-language outputs may reflect language-specific features rather than underlying semantics. To mitigate this dependence, we apply a *double translation* procedure: all justifications are translated into English and into Italian using the `deep-translator` library, embedded separately with LaBSE, and then averaged at the vector level. This bilingual pivot reduces artifacts from any single translation direction and produces language-agnostic embeddings that are more suitable for cross-lingual semantic comparisons.

Error profiling

In addition to stance and reasoning analysis, we track robustness indicators by summarizing JSON parsing failures, soft refusals, and empty outputs per model and condition. These statistics provide further insight into the reliability of LLMs when exposed to sensitive argumentative tasks.

Chapter 4

Analysis of Results

4.1 Dataset Exploration

Distinct conclusions. The dataset contains **332** distinct conclusions. The number of arguments per conclusion is highly variable, ranging from as few as **1** to as many as **114**. This indicates that the dataset is not well balanced across different conclusions.

Stance distribution. Figure 4.1 reports the overall distribution of stance labels relative to the conclusion. The corpus is moderately balanced, with “in favor of” slightly more common.

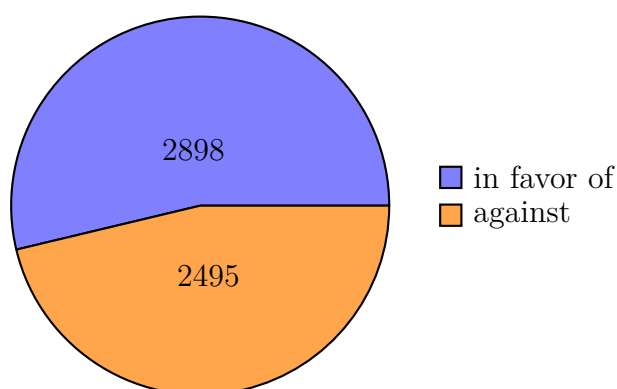


Figure 4.1: Distribution of stance labels relative to conclusions.

4.2 Topic Modeling Results

After applying topic modeling with **BERTopic**, the output is stored in a CSV file where each row corresponds to a discovered cluster. Each row contains the following fields: **Topic** (cluster ID), **Count** (number of arguments assigned to that cluster), **Name** (an auto-generated label from the most salient keywords), **Representation** (the highest c-TF-IDF keywords), and **Representative_Docs** (sample premises from the cluster).

Table 4.1 reports several clusters that were selected as the topics of interest for this project, including nuclear weapons, sex selection, prostitution, economic sanctions, multi-party systems, atheism, compulsory voting, and libertarianism.

Figure 4.2 illustrates the top seven most frequent clusters together with their representative keywords and their relative importance scores (as assigned by BERTopic). This provides intuition about the main lexical features that characterize each cluster.

Finally, Figure 4.3 shows the *intertopic distance map*, a two-dimensional visualization based on reduced embeddings. Clusters that appear close to each other share overlapping semantic content, whereas clusters that are well separated correspond to more distinct themes. Some overlap is visible, which indicates related arguments across clusters, while others are clearly distinct.

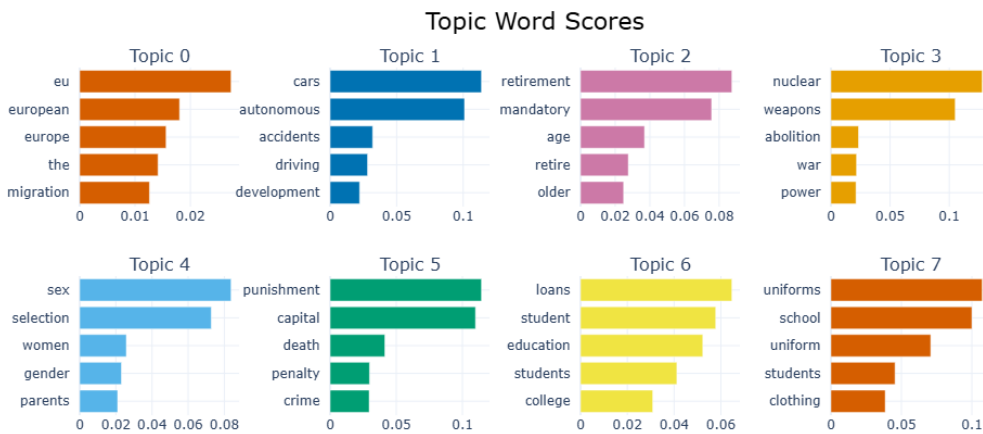


Figure 4.2: Topic word scores for selected clusters (higher bars indicate higher c-TF-IDF weight).



Figure 4.3: Intertopic distance map from BERTopic showing relative topic positions and sizes.

Topic	Count	Name	Representation (top keywords)
4	114	Sex selection / women / gender	sex, selection, women, gender, parents, child, family, men, legalize, rape
3	117	Nuclear weapons / abolition / war	nuclear, weapons, abolition, war, power, energy, destruction, peace, world, fight
17	98	Prostitution / legalization	prostitution, legalizing, women, prostitutes, legalize, sex, profession, legalising, safer, trafficking
15	100	Economic sanctions / countries	sanctions, economic, countries, end, china, effective, country, hurt, citizens, trade
18	98	Multi-party political systems	party, multi, system, parties, political, systems, two, more, would, rti
22	89	Atheism / religion / belief	atheism, adopt, religion, believe, wars, religious, adopted, god, belief, beliefs
37	47	Compulsory voting / turnout	voting, compulsory, voter, turnout, vote, would, representative, election, voice, elections
40	42	Libertarianism / adopt / government	libertarianism, adopt, would, government, adopted, we, chaos, because, should, anarchy

Table 4.1: Selected BERTopic clusters: counts, readable names, and top c-TF-IDF keywords.

4.3 Agreement Ratio Analysis

In this section we report the agreement ratios for each of the eight selected topics, across models (GPT, LLaMA) and prompting conditions (role-based, native). Each figure shows how often the model’s stance matched the gold stance for each country role, including the neutral assistant baseline.

General patterns.

- The **assistant baseline** is generally stable across topics and often lies in the mid-range of agreement ratios.
- **GPT** tends to be more consistent across roles and languages, with smaller gaps between conditions.
- **LLaMA** shows larger variability: in some topics, its agreement ratios diverge widely across roles or between role-based and native prompts.
- The magnitude of the **role vs. native difference** is topic-dependent: for some (e.g., prostitution, multi-party system) the results are close, while for others (e.g.,

sanctions, atheism) the divergence is large.

Topic-wise observations.

- **No nuclear weapons**

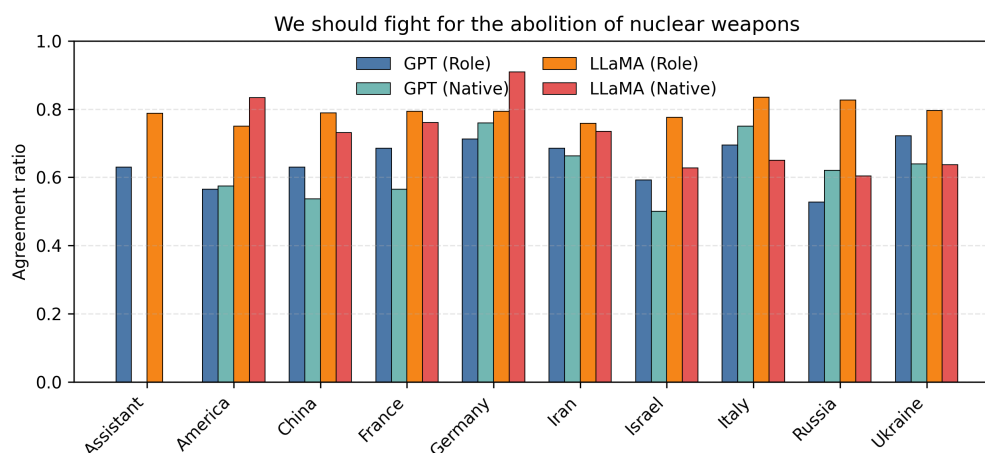


Figure 4.4: Agreement ratios by country for the topic “No nuclear weapons.”

(Figure 4.4): GPT produces agreement ratios around 0.6–0.8 with limited role/native spread. LLaMA is more variable, especially for Iran and Germany. Role vs. native differences are moderate.

- **Legalize sex selection**

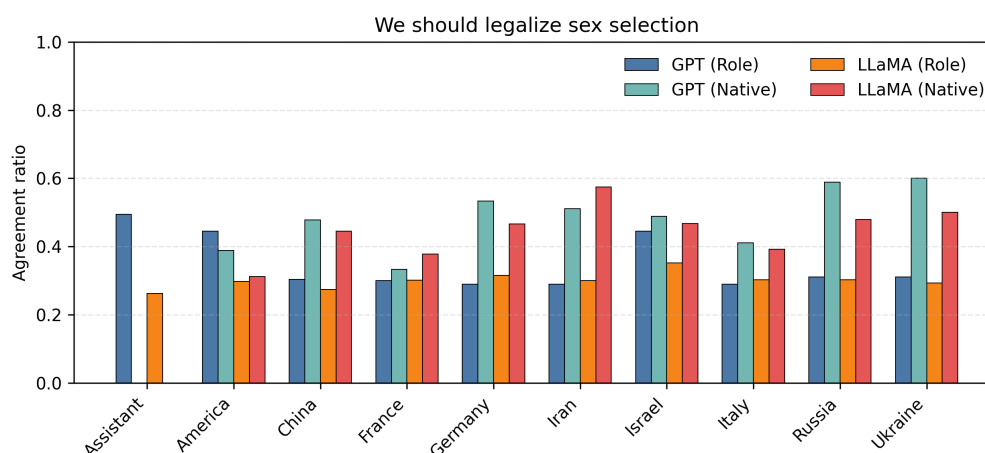


Figure 4.5: Agreement ratios by country for the topic “Legalize sex selection.”

(Figure 4.5): Both models lean toward disagreement (ratios often below 0.5). Larger role/native gaps appear in some countries (e.g., Iran, China).

- **No economic sanctions**

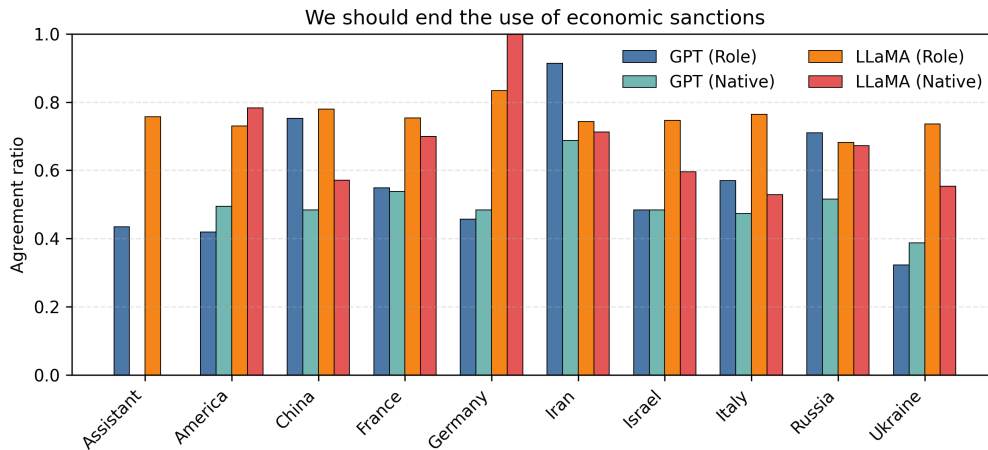


Figure 4.6: Agreement ratios by country for the topic “No economic sanctions.”

(Figure 4.6): Strong divergences. GPT remains in the 0.4–0.7 range, while LLaMA peaks close to 0.9 for Iran (role-based). Role/native gaps are particularly large in LLaMA.

- **Legalize prostitution**

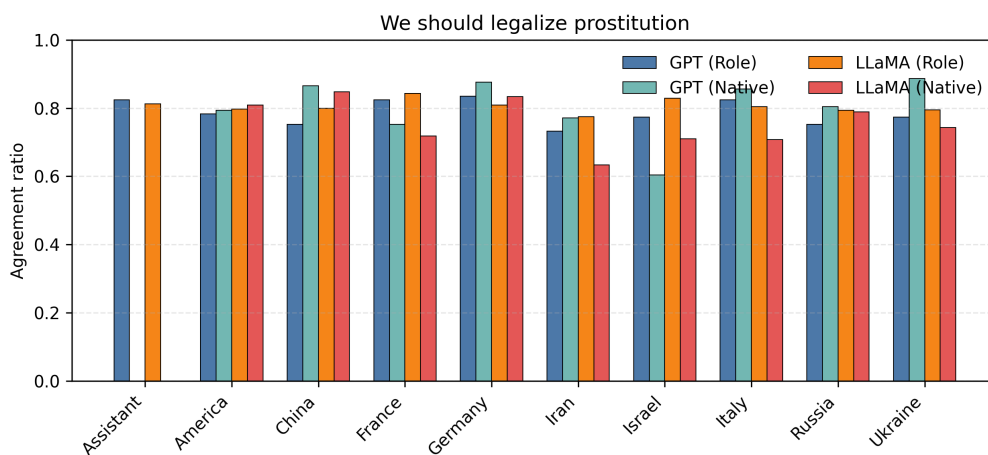


Figure 4.7: Agreement ratios by country for the topic “Legalize prostitution.”

(Figure 4.7): High agreement (> 0.7) across all models and conditions. Role vs. native differences are minimal.

- **Adopt multi-party system**

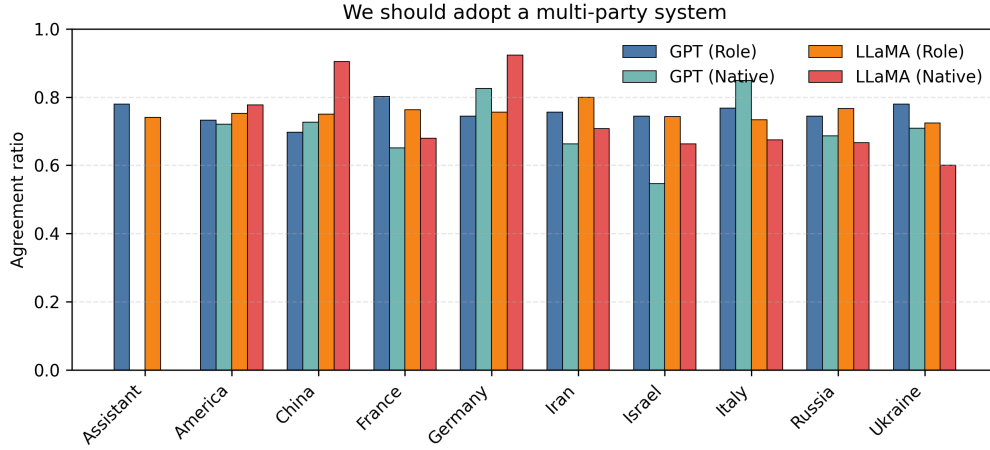


Figure 4.8: Agreement ratios by country for the topic “Adopt multi-party system.”

(Figure 4.8): Both models yield high agreement (0.7–0.9). Role and native completions are closely aligned, with little divergence.

- **Adopt atheism**

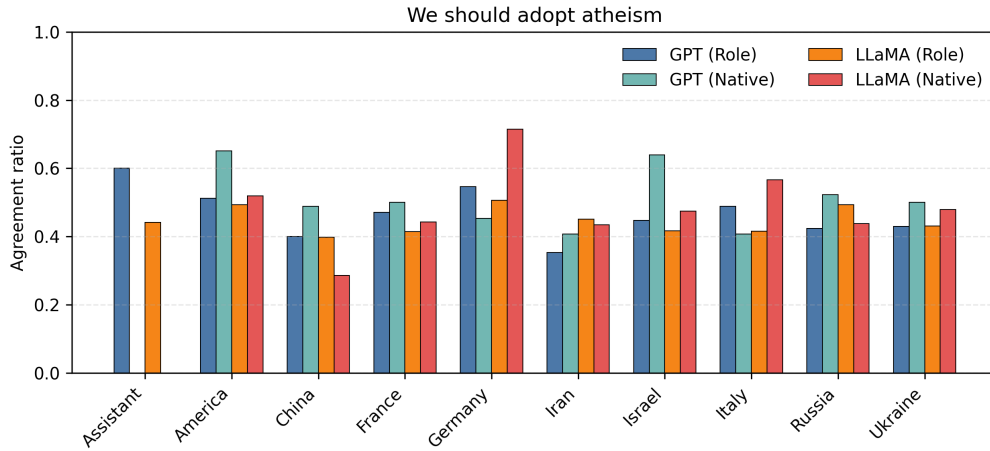


Figure 4.9: Agreement ratios by country for the topic “Adopt atheism.”

(Figure 4.9): Strong variation across roles and models. GPT tends to remain below 0.6, while LLaMA shows higher spread and larger role/native gaps. Divergences are particularly marked in countries with religious framing.

- **Compulsory voting**

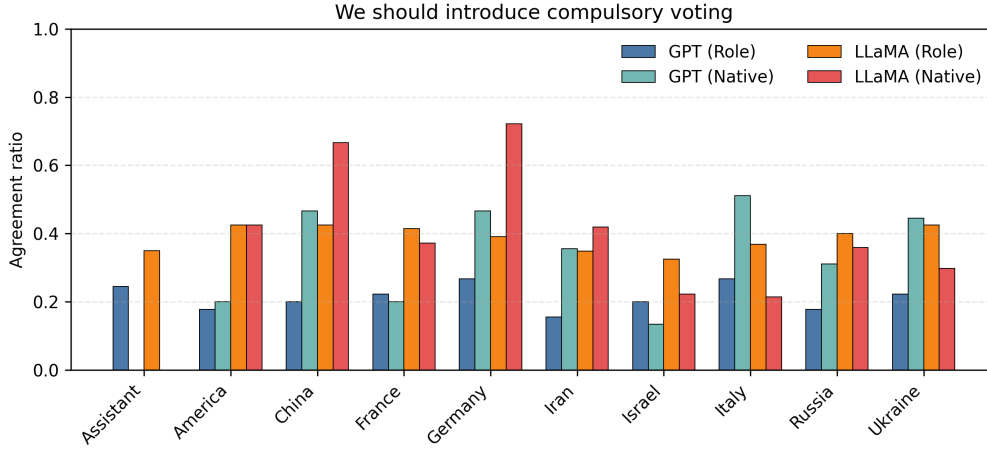


Figure 4.10: Agreement ratios by country for the topic “Compulsory voting.”

(Figure 4.10): The lowest agreement ratios overall. GPT mostly falls around 0.2–0.4, while LLaMA ranges more widely, with some roles exceeding 0.6. Role/native divergence is higher than in most other topics.

- **Adopt libertarianism**

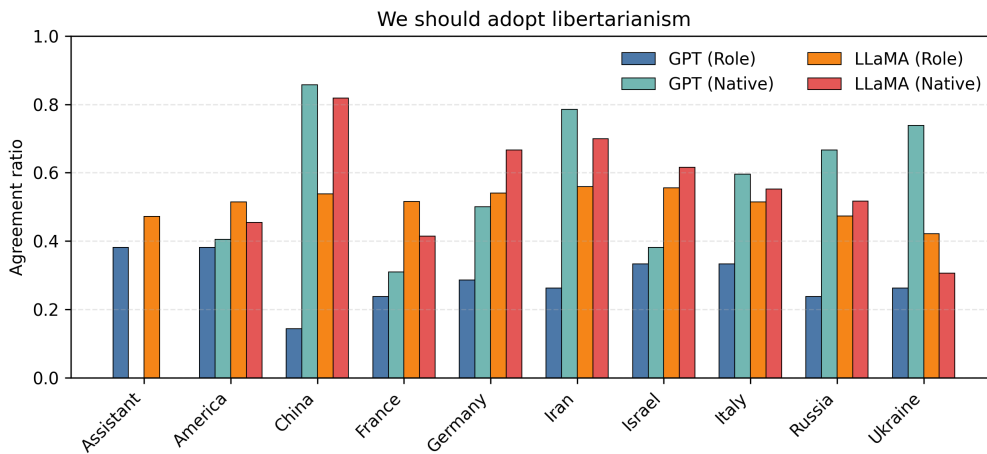


Figure 4.11: Agreement ratios by country for the topic “Adopt libertarianism.”

(Figure 4.11): Mid-range ratios (0.4–0.7). Both models show noticeable differences between role and native completions, though the size of the gap varies across countries.

Summary. Across topics, GPT demonstrates more stable agreement ratios with smaller role/native differences, while LLaMA exhibits stronger variability. Topics such as *pros-*

titution and *multi-party system* are consistently aligned across conditions, whereas *sanc-tions*, *atheism*, and *compulsory voting* show the largest divergences.

4.4 Embedding Similarity Results

In this section, we analyze the embedding similarities between conditions:

1. Role-based vs. Assistant responses,
2. Native vs. Role-based responses,
3. Native vs. Assistant responses.

Each bar represents the mean cosine similarity across topics for a given country, with error bars denoting the standard deviation.

A **higher similarity value** indicates that the two conditions being compared (e.g., role-based vs. assistant) produced embeddings that were closer in semantic space, suggesting that the model generated more similar reasoning structures across those prompting conditions. Conversely, a **lower similarity** means that the responses diverged more strongly in their semantic content, implying that the model adapted its reasoning depending on the prompt type.

A **small standard deviation** indicates that the model’s behavior is stable across all topics, i.e., it shows a consistent degree of similarity between the compared conditions. A **large standard deviation**, on the other hand, implies greater variability: for some topics, the model produced very similar responses across conditions, while for others, the responses diverged more significantly. This variability is particularly informative as it points to topic-dependent sensitivity in the model’s outputs.

4.4.1 GPT-4o-mini

Figure 4.12 shows that similarities between role-based and assistant responses remain relatively low across countries, generally centered around 0.55–0.6 with large standard deviations. Figure 4.13 highlights that native vs. role-based similarities are consistently

higher, especially for America, China, and Italy, where scores approach 0.7. Figure 4.14 indicates that native vs. assistant similarities are the lowest of the three comparisons, with averages often near 0.5, suggesting stronger divergence between native-language completions and assistant completions.

4.4.2 LLaMA 3.1-70B

For LLaMA, results appear different. In Figure 4.15, role-based vs. assistant similarities are moderate (around 0.6–0.65), with relatively wide error bars for some countries (e.g., Iran, Ukraine). Figure 4.16 shows consistently higher similarities (0.65–0.8), particularly for America and France, which exhibit closer alignment between native and role-based answers. Finally, Figure 4.17 reveals that native vs. assistant similarities are comparable to the role-based vs. assistant case, again hovering near 0.65 for most countries.

4.4.3 General Patterns

Two clear trends emerge:

- Native vs. Role-based similarities are generally the highest across both models, suggesting these two prompting strategies produce the most consistent outputs.
- Native vs. Assistant similarities tend to be lowest, implying that the “helpful assistant” completions diverge more strongly from role-based or native completions.

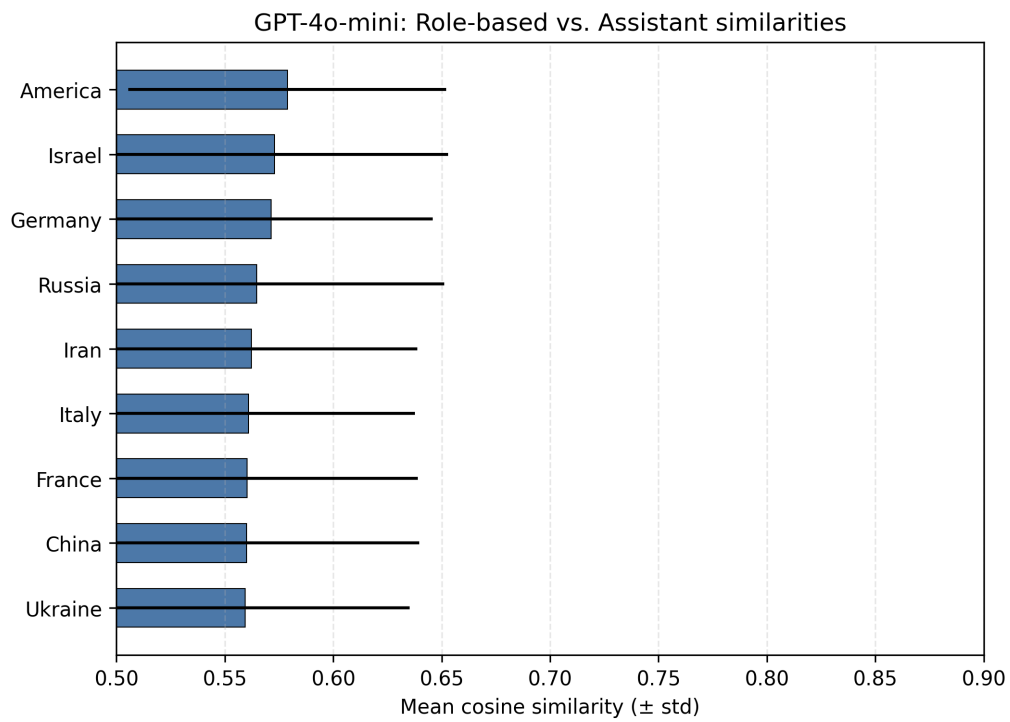


Figure 4.12: GPT-4o-mini: Role-based vs. Assistant similarities.

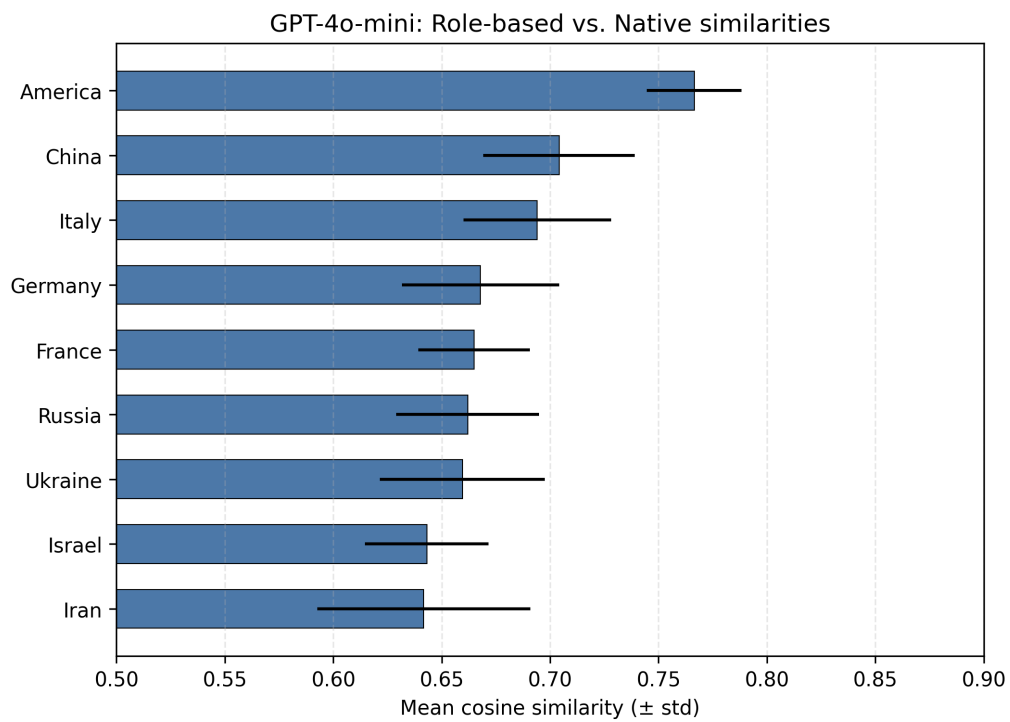


Figure 4.13: GPT-4o-mini: Native vs. Role-based similarities.

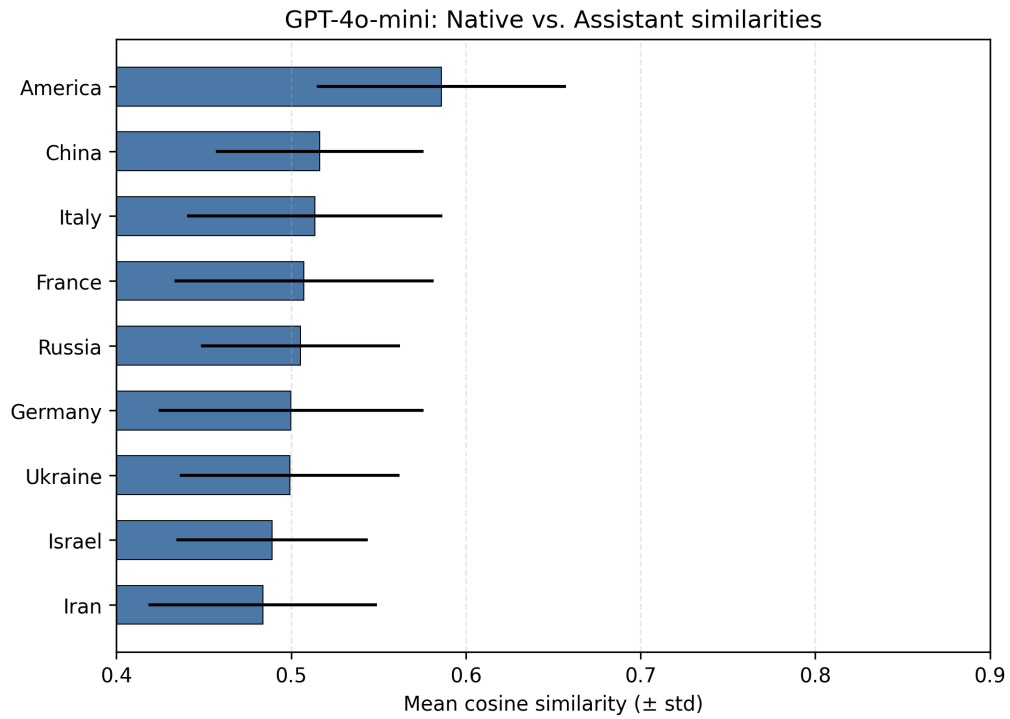


Figure 4.14: GPT-4o-mini: Native vs. Assistant similarities.

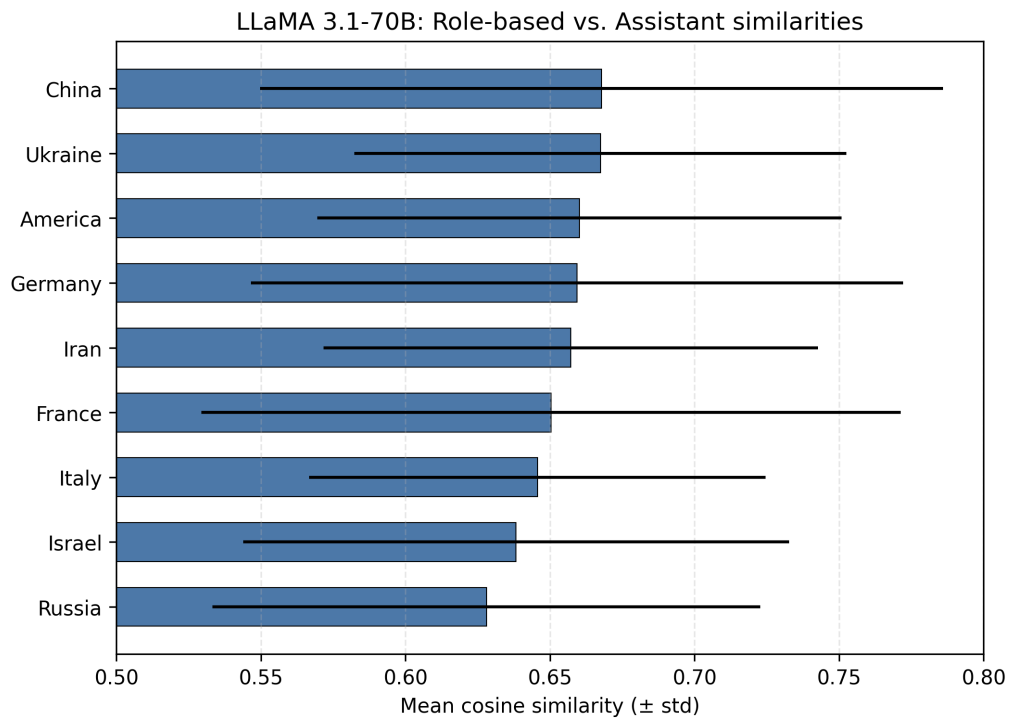


Figure 4.15: LLaMA 3.1-70B: Role-based vs. Assistant similarities.

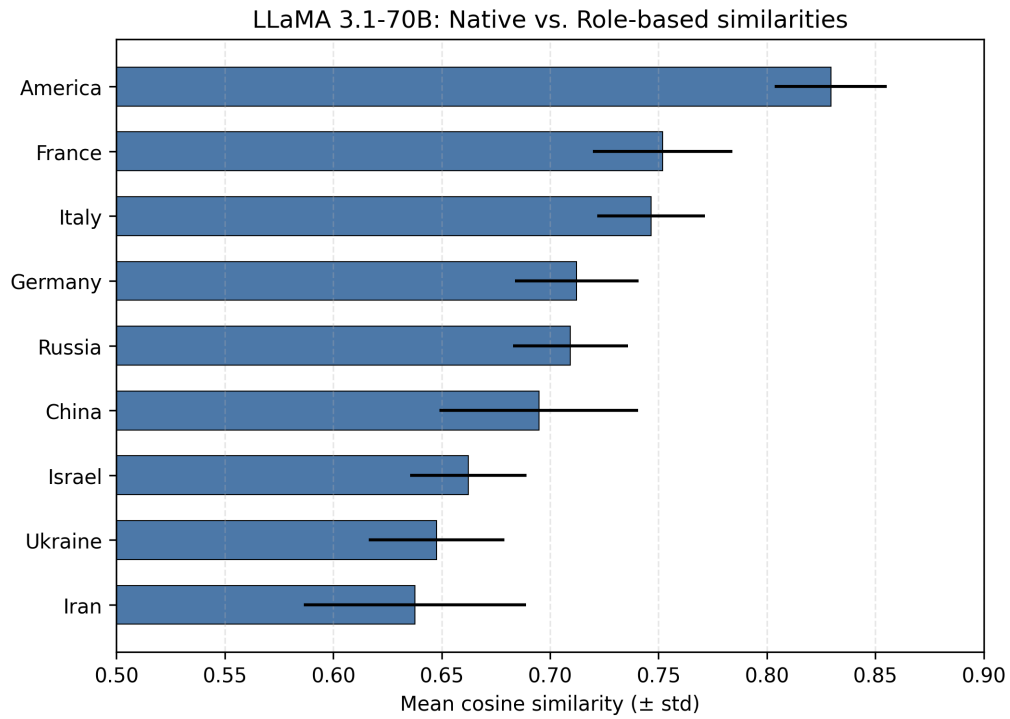


Figure 4.16: LLaMA 3.1-70B: Native vs. Role-based similarities.

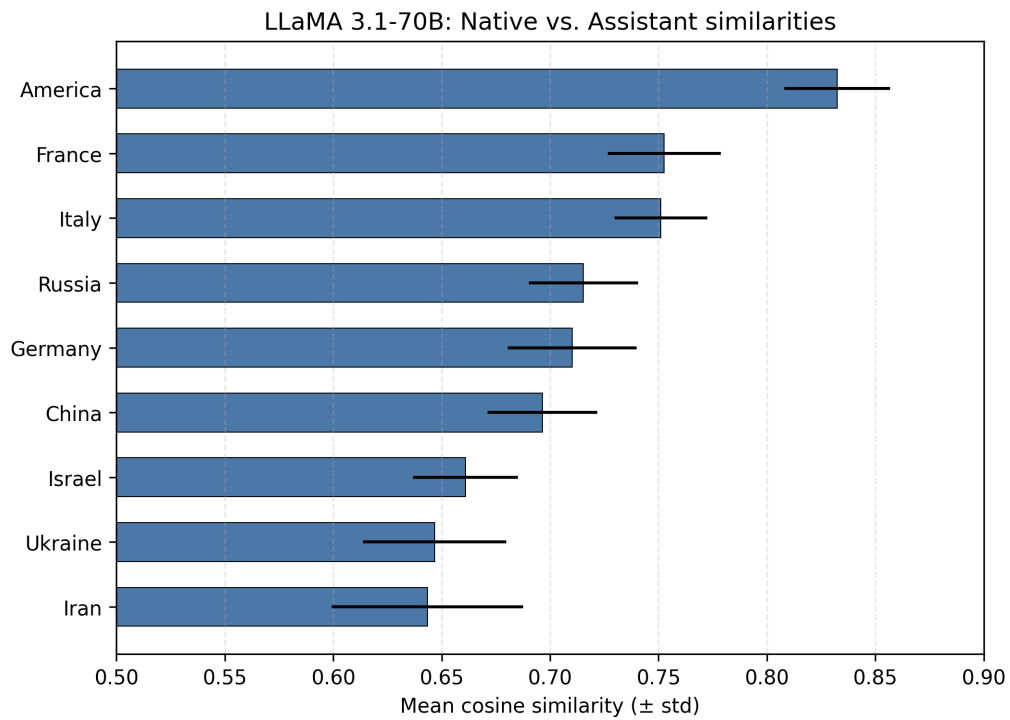


Figure 4.17: LLaMA 3.1-70B: Native vs. Assistant similarities.

4.5 Model Comparison (GPT vs LLaMA)

Chapter 5

Limitations and Future works

References

- [1] OpenAI et al. *GPT-4 Technical Report*. 2023. arXiv: 2303.08774 [cs.CL]. URL: <https://arxiv.org/abs/2303.08774>.
- [2] Hugo Touvron et al. *LLaMA: Open and Efficient Foundation Language Models*. 2023. arXiv: 2302.13971 [cs.CL]. URL: <https://arxiv.org/abs/2302.13971>.
- [3] Laura Weidinger et al. *Ethical and Social Risks of Harm from Language Models*. 2021. arXiv: 2112.04359 [cs.CL]. URL: <https://arxiv.org/abs/2112.04359>.
- [4] Shibani Santurkar et al. “Whose Opinions Do Language Models Reflect?” In: *Proceedings of the 40th International Conference on Machine Learning*. Ed. by Andreas Krause et al. Vol. 202. Proceedings of Machine Learning Research. PMLR, 2023, pp. 29971–30004. URL: <https://proceedings.mlr.press/v202/santurkar23a.html>.
- [5] Junfeng Jiao et al. *Navigating LLM Ethics: Advancements, Challenges, and Future Directions*. 2025. arXiv: 2406.18841 [cs.CY]. URL: <https://arxiv.org/abs/2406.18841>.
- [6] Keita Kurita et al. *Measuring Bias in Contextualized Word Representations*. 2019. arXiv: 1906.07337 [cs.CL]. URL: <https://arxiv.org/abs/1906.07337>.
- [7] Moin Nadeem, Anna Bethke, and Siva Reddy. *StereoSet: Measuring stereotypical bias in pretrained language models*. 2020. arXiv: 2004.09456 [cs.CL]. URL: <https://arxiv.org/abs/2004.09456>.

- [8] Md Abdul Aowal et al. *Detecting Natural Language Biases with Prompt-based Learning*. 2023. arXiv: 2309.05227 [cs.CL]. URL: <https://arxiv.org/abs/2309.05227>.
- [9] Nicole Gross. “What ChatGPT Tells Us about Gender: A Cautionary Tale about Performativity and Gender Biases in AI”. In: *Social Sciences* 12.8 (2023). ISSN: 2076-0760. DOI: 10.3390/socsci12080435. URL: <https://www.mdpi.com/2076-0760/12/8/435>.
- [10] Aleksandra Urman and Mykola Makhortykh. “The silence of the LLMs: Cross-lingual analysis of guardrail-related political bias and false information prevalence in ChatGPT, Google Bard (Gemini), and Bing Chat”. In: *Telematics and Informatics* 96 (2025), p. 102211. ISSN: 0736-5853. DOI: <https://doi.org/10.1016/j.tele.2024.102211>. URL: <https://www.sciencedirect.com/science/article/pii/S0736585324001151>.
- [11] Pranav Narayanan Venkit et al. *Nationality Bias in Text Generation*. 2023. arXiv: 2302.02463 [cs.CL]. URL: <https://arxiv.org/abs/2302.02463>.
- [12] Richard Rogers and Xiaoke Zhang. “The Russia–Ukraine War in Chinese Social Media: LLM Analysis Yields a Bias Toward Neutrality”. In: *Social Media + Society* 10 (May 2024). DOI: 10.1177/20563051241254379.
- [13] Tomas Mikolov et al. *Efficient Estimation of Word Representations in Vector Space*. 2013. arXiv: 1301.3781 [cs.CL]. URL: <https://arxiv.org/abs/1301.3781>.
- [14] Jeffrey Pennington, Richard Socher, and Christopher Manning. “GloVe: Global Vectors for Word Representation”. In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Ed. by Alessandro Moschitti, Bo Pang, and Walter Daelemans. Doha, Qatar: Association for Computational Linguistics, Oct. 2014, pp. 1532–1543. DOI: 10.3115/v1/D14-1162. URL: <https://aclanthology.org/D14-1162/>.
- [15] Matthew E. Peters et al. *Deep contextualized word representations*. 2018. arXiv: 1802.05365 [cs.CL]. URL: <https://arxiv.org/abs/1802.05365>.

- [16] Jacob Devlin et al. *BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding*. 2019. arXiv: 1810.04805 [cs.CL]. URL: <https://arxiv.org/abs/1810.04805>.
- [17] Nils Reimers and Iryna Gurevych. *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks*. 2019. arXiv: 1908.10084 [cs.CL]. URL: <https://arxiv.org/abs/1908.10084>.
- [18] Fangxiaoyu Feng et al. *Language-agnostic BERT Sentence Embedding*. 2022. arXiv: 2007.01852 [cs.CL]. URL: <https://arxiv.org/abs/2007.01852>.
- [19] Ashish Vaswani et al. *Attention Is All You Need*. 2023. arXiv: 1706.03762 [cs.CL]. URL: <https://arxiv.org/abs/1706.03762>.
- [20] Alec Radford et al. *Improving language understanding by generative pre-training*. 2018.
- [21] Tom B. Brown et al. *Language Models are Few-Shot Learners*. 2020. arXiv: 2005.14165 [cs.CL]. URL: <https://arxiv.org/abs/2005.14165>.
- [22] Long Ouyang et al. *Training language models to follow instructions with human feedback*. 2022. arXiv: 2203.02155 [cs.CL]. URL: <https://arxiv.org/abs/2203.02155>.
- [23] Bernd Carsten Stahl and Damian Eke. *The ethics of ChatGPT – Exploring the ethical issues of an emerging technology*. 2024. DOI: <https://doi.org/10.1016/j.ijinfomgt.2023.102700>. URL: <https://www.sciencedirect.com/science/article/pii/S0268401223000816>.
- [24] Nikita Nangia et al. *CrowS-Pairs: A Challenge Dataset for Measuring Social Biases in Masked Language Models*. 2020. arXiv: 2010.00133 [cs.CL]. URL: <https://arxiv.org/abs/2010.00133>.
- [25] Patrick Haller, Ansar Aynedinov, and Alan Akbik. *OpinionGPT: Modelling Explicit Biases in Instruction-Tuned LLMs*. 2023. arXiv: 2309.03876 [cs.CL]. URL: <https://arxiv.org/abs/2309.03876>.

- [26] Anthony G. Greenwald, Debbie E. McGhee, and Jordan L. K. Schwartz. “Measuring Individual Differences in Implicit Cognition: The Implicit Association Test”. In: *Journal of Personality and Social Psychology* 74.6 (1998), pp. 1464–1480. DOI: 10.1037/0022-3514.74.6.1464. URL: <https://psycnet.apa.org/doi/10.1037/0022-3514.74.6.1464>.
- [27] Chandler May et al. *On Measuring Social Biases in Sentence Encoders*. 2019. arXiv: 1903.10561 [cs.CL]. URL: <https://arxiv.org/abs/1903.10561>.
- [28] Nailia Mirzakhmedova et al. *The Touché23-ValueEval Dataset for Identifying Human Values behind Arguments*. 2023. arXiv: 2301.13771 [cs.CL]. URL: <https://arxiv.org/abs/2301.13771>.
- [29] Nikita Soni et al. “Human Language Modeling”. In: *Findings of the Association for Computational Linguistics: ACL 2022*. Ed. by Smaranda Muresan, Preslav Nakov, and Aline Villavicencio. Dublin, Ireland: Association for Computational Linguistics, May 2022, pp. 622–636. DOI: 10.18653/v1/2022.findings-acl.52. URL: <https://aclanthology.org/2022.findings-acl.52/>.
- [30] Maarten Grootendorst. *BERTopic: Neural topic modeling with a class-based TF-IDF procedure*. 2022. arXiv: 2203.05794 [cs.CL]. URL: <https://arxiv.org/abs/2203.05794>.
- [31] Nidhal Baccouri. *deep-translator: A flexible free and unlimited python tool to translate between different languages in a simple way using multiple translators*. <https://pypi.org/project/deep-translator/>. Version 1.11.4, accessed: 2025-08-24. 2020.
- [32] OpenAI. *Learn how to use JSON mode*. <https://platform.openai.com/docs/guides/json-mode>. Accessed: 2025-07-02. 2025.